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TOPICAL REVIEW

MCQs Generation With Large Language Models: A Survey of Methodologies, Evolution, and Open Research Issues

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ABSTRACT In recent years, there has been growing interest in the automatic generation of Multiple Choice Questions (MCQs), mainly to reduce the effort and time involved in manual question construction. Many researchers and practitioners have tried different methods to produce accurate and good-quality MCQs automatically. The progress in this area has been steady but also diverse, which makes it difficult to identify which approaches work best in practice. The research trend has moved gradually from early rule-based methods to more advanced systems based on Large Language Models (LLMs). Although several review papers and surveys have summarized earlier work, most of them give limited discussion on how the field has evolved over time or what key problems remain to be solved. This paper aims to fill that gap by reviewing important studies on MCQs generation and providing a structured overview of the main question types, their essential components, and the processes involved in automated generation. It also discusses the essential stages in a general system architecture, including preprocessing, key selection, and distractor generation. In addition, the review summarizes the most frequently used datasets and evaluation measures for assessing question quality and highlights important applications and research challenges. In doing so, it outlines possible future directions to advance the use of automated MCQs generation in education, training, and professional assessment.

INDEX TERMS Assessments, multiple choice questions (MCQs), large language models (LLMs), automatic MCQs generation, generative AI in education, educational assessment.

I. INTRODUCTION

Assessment is a vital part of the learning process, and among the different forms of assessment, the MCQs remains one of the most commonly used because of its simplicity, efficiency, and objectivity [1], [2], [3], [8], [9], [10], [18]. A typical MCQs item consists of a question stem, one correct option known as the key, and several distractors that are incorrect yet plausible enough to challenge a learner's understanding. Due to their clear structure and ease of use at scale, MCQs are extensively applied in educational testing, Massive Open Online Courses (MOOCs), and various professional certification programs [6], [48]. Their objective nature and suitability for computer-based systems has led to widespread adoption in standardized testing (e.g., Scholastic Assessment Test (SAT), Graduate Record Examination (GRE)) and across several domains and languages [4], [34], [97], [131], [151], [158]. However, the manual creation of high-quality MCQs is a significant challenge for test creators, as it is laborious, time-consuming and requires strong domain and

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TABLE 1. Comparison with various relevant review papers.

Paper	Focus	No. of Studies	Search Strategy	Key Finding
[1]	Automatic MCQs generation from text	86	Journal articles, criteria-based selection	Generic 6-phase workflow, techniques per phase, identified challenges
[2]	Automated medical case-based MCQs generation	98	PubMed, Scopus, Web of Science	Methodologies, applications, evaluation techniques, and future research directions for medical case-based MCQs generation are discussed.
[8]	Techniques for development of MCQs generation from text	83	Sequential and random search strategy	Discusses precise methods at different stages of MCQs generation and suggests areas for improvement in quality
[9]	Automatic question generation (AQG) for educational purposes	72	5 databases, AIED venues, snowballing	Articulates trends, advances, and changes in AQG, Suggests areas for improvement and future possibilities.
[10]	LLMs for generating medical MCQs	8	MEDLINE	LLMs can produce competent questions and validity evidence varies across studies
[101]	Medical case-based MCQs generation	38	Examining the development of medical case-based MCQs	Discusses history, design ideas, methods, challenges validity, reliability.
[185]	ChatGPT prompts for medical MCQs generation and validity	23	PubMed and Web of Science	Prompts for ChatGPT varied, validity evidence for Artificial Intelligence (AI)-generated MCQs is unclear, highlighting need for further research
This Paper	Automatic MCQs generation across fields	106	Detailed survey methodology, mentions queries	Discusses evolution from rule-based to LLMs classification, generic architecture, evaluation methods, applications and future work

pedagogical expertise to ensure accuracy, relevance, and balanced difficulty while avoiding ambiguity and bias [1], [49], [69].

The laborious nature of manual creation was the core reason behind the research towards the process to automate the generation of MCQs [1], [8], [9], [18]. At the beginning, foundation was laid in 1997 when David Conium developed a rule-based system that relied on predefined templates. However, the system shows limitations with reference to its ability to generate contextually relevant and linguistically complex questions [65]. The development of the Transformer architecture in 2017 marked a paradigm shift, which catalysed the development of powerful LLMs such as Bidirectional Encoder Representations from Transformers (BERT), Bidirectional and Auto-Regressive Transformers (BART), Generative Pre-trained Transformer (GPT), and Text-to-Text Transfer Transformer (T5) [7], [42], [54], [58], [67], [172]. These huge models transformed natural language generation by taking advantage of massive datasets to reveal complex linguistic patterns. As a result, the approach can now produce high-quality, diverse, and contextually aware MCQs with minimal human intervention [4], [18], [43], [66], [186].

Evaluating the quality of automatically generated questions is not straightforward and needs more than one type of assessment. In most studies, automatic measures such as Bilingual Evaluation Understudy (BLEU), Recall-Oriented Understudy for Gisting Evaluation (ROUGE), Recall-Oriented Understudy for Gisting Evaluation-Longest Common Subsequence (ROUGE-L), and Item Discrimination (ID) are used to check how close a generated question is to a human-written one [18], [66]. These measures give some numerical idea of similarity, but they do not really tell whether a question is educationally useful or conceptually sound. For that reason, human evaluation is still necessary, especially to see if the stem is clear, the key is correct, and the distractors make sense [4], [66], [186]. In some recent works, it is also shown that the systems can be adapted to specific fields by using domain data or by adding Retrieval-Augmented Generation (RAG) techniques. This type of adaptation has proved to be more effective in specialized areas such as medicine and engineering, where correctness and precision are very important [10], [48], [66], [67], [101], [139], [176], [181].

Earlier surveys have given a general view of progress in this area and discussed how different methods have developed over time, as summarized in Table 1. The first group of reviews mostly described basic, multi-step workflows for AQG from text [1], [8], [9]. Later studies started focusing on specific domains, for example, generating medical case questions or using LLMs models such as ChatGPT [2], [10], [101], [185]. These reviews have shown that AI-based systems can perform well, but they also point out that there is still no clear agreement on how to measure the quality and reliability of the generated questions [181].

This paper attempts to address this gap by providing a new and complete review of how question generation has evolved from the early rule-based methods to the latest LLMs-based ones by covering different application areas. The goal is to combine the existing work into one consistent overview and reflect the fast development in this field.

To guide the study, we consider the following questions:

1) How have MCQs generation methods developed from simple rule-based systems to modern LLMs-based techniques?
2) What are the main strategies used by LLMs systems to generate good-quality MCQs automatically?
3) What open problems and opportunities remain in this area of LLMs-based MCQs generation?

By answering these questions, this paper organizes the available literature, identifies gaps in current work, and suggests possible research directions. The discussion follows

an evolutionary view, showing how automated question generation has moved from basic rule-driven designs to advanced LLMs-powered systems.

The main contributions of this survey can be summarized as:

- Describing the development of MCQs generation systems, from early rule-based models to modern LLMs-based approaches.
- Explaining the theoretical background and structure of MCQs, and their use in different domains.
- Providing a simple classification of existing techniques, such as rule-based, template-driven, ontology-based, Machine Learning (ML)/Deep Learning (DL), and LLMs approaches.
- Presenting a common framework that includes the main stages of MCQs generation, including pre-processing, key selection, and Distractor Generation (DG).
- Reviewing different evaluation approaches, from automatic metrics to human and psychometric analysis, with their respective advantages and weaknesses.
- Listing key datasets used for training and fine-tuning, with details about their size, coverage, and availability.
- Discussing how MCQs are applied in various fields such as education, language learning, medicine, and professional testing.
- Highlighting the strengths and weaknesses of LLMs-based systems, including issues like bias, distractor quality, and scalability.
- Pointing out future challenges and opportunities, such as adaptive learning, multi-modal question generation, ethical considerations, and hybrid human-AI collaboration.

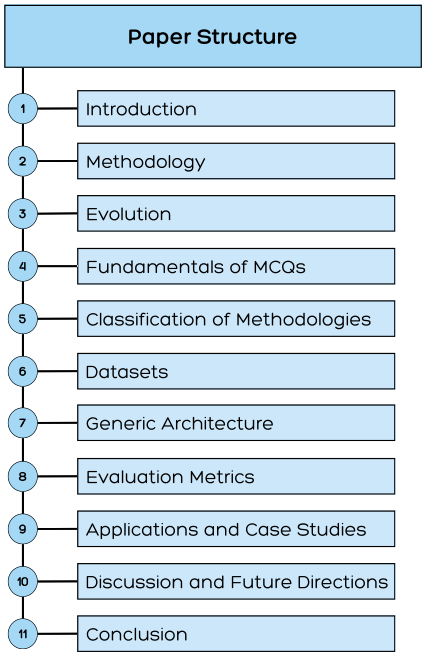


FIGURE 1. Paper structure — overview of the main contents of this survey.

Figure 1 gives a short view of how the paper is organized. Section I introduces the motivation behind automated MCQs generation. Section II explains the process used for literature selection and analysis. Section III outlines the historical development of assessment and question generation methods, from early rule-based designs to the present LLMs-based stage. Section IV briefly introduces the basic parts of an MCQs. Section V describes different generation methods, followed by Section VI, which summarizes key datasets used for training and testing. Section VII presents a common system architecture, while Section VIII discusses the evaluation metrics. Section IX focuses on major applications, and Section X discusses the key challenges and research directions. Finally, Section XI concludes the paper with a short summary.

II. METHODOLOGY

This survey employed a systematic literature review methodology to identify, screen, and analyze the literature on automatic MCQs generation. The process, illustrated in Figure 2, was designed to provide a comprehensive and unbiased overview of the field's evolution, dominant techniques, and open challenges.

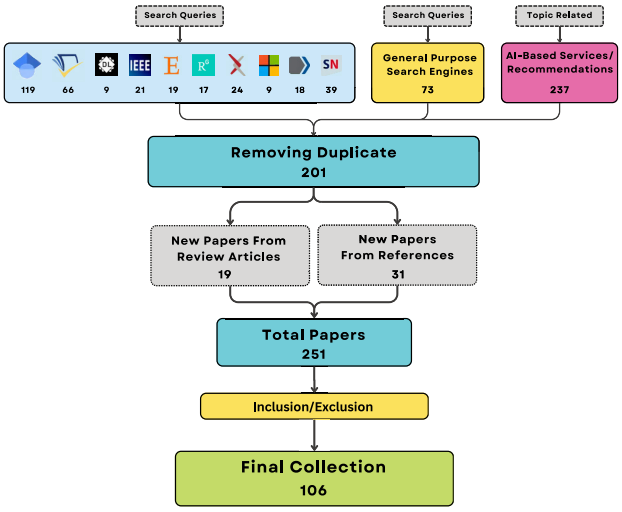
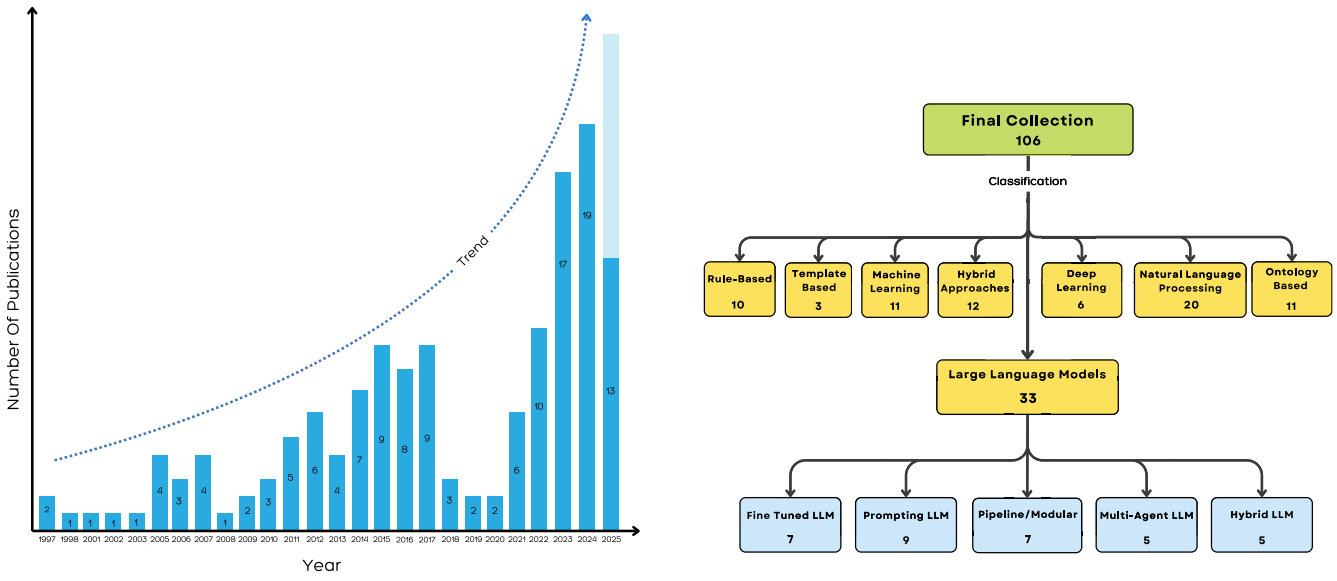


FIGURE 2. Survey methodology: A visual flow of applied methodology for paper collection of MCQs generation approaches.

A. SEARCH AND SELECTION OF STUDIES

We conducted a systematic search of major academic libraries, including ACM Digital Library, IEEE Xplore, Google Scholar, Scopus, and Semantic Scholar, for studies published between 1997 and 2025. The search query was constructed using keywords and their synonyms across three main categories:

- **Primary Concept**: (“automatic question generation”, “MCQs generation”, “multiple choice question generation”, “assessment item generation”)
- **Technology/Method**: (“LLM”, “large language model”, “transformer”, “BERT”, “GPT”, “neural network”, “rule-based”, “ontology”)



(a) Number of selected publications on automatic MCQs generation per year, showing the general trend.

(b) Number of publications in each methodology.

FIGURE 3. Publication statistics: (a) yearly trends and (b) classification based on methodology.

- **Application Domain**: (“education”, “e-learning”, “automated assessment”, “pedagogy”)

These keywords were blended using Boolean operators (AND, OR) to form broad search strings, such as (“automatic question generation” OR “MCQs generation”) AND (“LLM” OR “GPT”), to capture literature encompassing from foundational rule-based systems to modern LLMs-based approaches. In order to make sure a wide-ranging coverage, this primary search was supplemented with a two-stage citation analysis (backward and forward search) [178] and recommendations from platforms like ResearchGate, Semantic Scholar and AI-Based systems.

The final collection of studies were then passed through a filtration process using a strict inclusion and exclusion criteria illustrated in Table 2, which underscore peer-reviewed, empirically evaluated using multiple available metrics, and novel contributions published in English. A total of 580 studies were selected from a top-level inspection that included only the title of the paper and the keywords used. These papers then underwent a careful process of scrutiny based on their abstracts. This process created a filter that facilitated the selection process, where from the initial pool of 580 studies collected from different sources, 201 duplicates were removed after carefully analyzing their context. After evaluating the remaining 379 articles by reviewing abstract, introduction, and methodology section with meticulous content skimming, a final set of 106 publications was chosen for comprehensive analysis.

TABLE 2. Inclusion and exclusion criteria for study selection.

Inclusion Criteria	Exclusion Criteria
Studies from 1997 to 2025.	Only manual creation of MCQs.
Published in English.	Publications in other languages.
Focus on automated MCQs generation.	General assessment techniques.
From top peer-reviewed venues.	Inaccessible or not peer-reviewed.
Effectiveness empirically shown.	Duplicated or overlapping studies.
Introducing novel methods.	Lacking novelty or strong contribution.

B. DATA EXTRACTION AND THEMATIC ANALYSIS

We performed a highly optimized systematic data extraction process for each paper in order to gather key information, including the methodologies used, application domains, evaluation methods, key findings of the developed system, and limitations reported in the study. The data extracted from these papers was then verified again to remove factual errors and context mismatch. After verification process, all the extracted data was synthesized using Thematic Analysis and Categorization Schema [179] to identify major trends, patterns, and research gaps.

A statistical analysis was carried out to visualize the field’s advancement from initial publication to the current status. This includes the annual publication volume of selected studies on automatic MCQs generation as shown in Figure 3a and the frequency of different methodological approaches which can be seen in Figure 3b. Table 3 provides a high-level overview of these approaches. This

TABLE 3. Key approaches for MCQs generation.

Approach	Description
Rule-Based Systems	Generates questions using predefined rules.
Template-Based Systems	Uses fixed or customized question structures.
Statistical & ML Models	Analyzes data to generate questions based on patterns.
Ontology-Based Systems	Uses structured knowledge for complex concepts.
DL Models	Creates complex questions using models trained on large text.
Natural Language Processing (NLP)-Based Systems	Extracts content for Question Generation (QG) using NLP.
LLMs	Generates high-quality, contextual questions (e.g., GPT, BERT, Gemini).
Hybrid Approaches	Combines multiple techniques for better quality.

multidimensional analysis provides a comprehensive view of the research landscape, from its historical context to emerging directions in automatic MCQs generation.

C. LIMITATIONS OF THE SURVEY

This survey has extracted a wide range of methodologies, but still has a number of limitations. Firstly, the search strategy employed may not have captured all relevant publications, specifically non-English studies or those in less accessible journals. Secondly, the classification of papers into individual methodologies involves some sort of subjectivity, as many studies use hybrid approaches. Finally, the LLMs-based MCQs generation approach is evolving at an exceptionally rapid speed. This review provides a snapshot of the current landscape, but new advancements will quickly emerge.

III. EVOLUTION

The history of educational assessment has seen a gradual transition from subjective, essay-based evaluations to objective, scalable formats. Early written and oral tests were criticized for not being fair and for having scores that were not always match up. In response, teachers used MCQs as a more reliable and fair way to test students.

The foundation for modern objective testing was laid in the early 20th century. Thorndike and others researchers came up with new substitutes to ask questions, and Yerkes's Army Alpha and Beta tests showed that MCQs could be used in tests during World War I [3], [174]. McCall and Ruch are one of those researchers who further cultivated objective testing approaches by highlighting the importance for a clear question construction [31]. Thurstone introduced scaling methods for the first time in 1925, giving test analysis a psychometric basis and making MCQs an important part of standardized tests like the SAT, GRE, and GMAT [69], [177].

With the advent of computer age in the late 1980s, assessment methods were transformed with the arrival of Computer-Based Testing (CBT) systems and Computerized Adaptive Test (CAT) systems for assessment [177]. On the one hand, these technologies simplified the completion of tests and allowed the use of adaptive difficulty, but on the other hand, they created an unprecedented need to have high-quality and large banks of questions. Manual generation of MCQs was a major bottleneck, which highlighted the necessity for an automated system to generate quality MCQs.

A. THE EVOLUTION OF AUTOMATED GENERATION

David Coniam in 1997 became the first author to introduce a dedicated system for automatic generation of MCQs using a rule-based approach to build vocabulary cloze tests on textual content [65]. Early systems of this sort relied heavily on the predetermined linguistic rules and template structures. Such systems, though offering some degree of controllability, were often fragile, difficult to scale, and had enormous problems in the area of scalability and adapting to new areas of expertise [71], [77].

The development of NLP in the early 2000s meant the implementation of statistical methods. Scholars used term extraction and shallow parsing to automate the generation of questions. At the same time, others used domain specific corpora with the item response theory to generate questions that were contextually relevant and had plausible distractors [34]. Later studies adopted different approaches in order to improve the process of question generation. Ontology-based systems took advantage of structured knowledge hierarchies to give rise to conceptually coherent and semantically aligned questions [36], [37], [38]. At the same time, DL models, particularly Recurrent Neural Networks (RNNs) and Long Short Term Memory (LSTM), brought significant improvements in the fluency and simplicity of generated text by learning complex linguistic patterns on a large-scale corpus of textual data [39], [40], [44], [45], [169].

One of the most crucial moments in the development of NLP is that Vaswani introduced the Transformer architecture in 2017 [66]. The self-attention mechanism and concurrent processing ability of the transformer based model enabled the development of strong pretrained models like BERT, GPT and T5 that have significantly enhanced the capability and fine-tuning of automated content generation [54], [58], [67].

B. THE MODERN ERA OF LARGE LANGUAGE MODELS

The early 2020s marked the beginning of a new era due to the breakthrough in the field of LLMs such as ChatGPT, Pathways Language Model (PaLM), Gemini, and Llama [7], [10], [54], [58], [67], [172]. They have shown state-of-the-art performance in professional and academic exams and are capable of producing high quality, contextually relevant MCQs when presented with a wide variety of sources such as textbooks and video transcripts [14], [180]. Their key applications include identifying important concepts, formulating questions aligned with learning objectives, and creating plausible distractors [42]. Moreover, their multilingual capabilities have increased the availability of automated assessments tools worldwide [180].

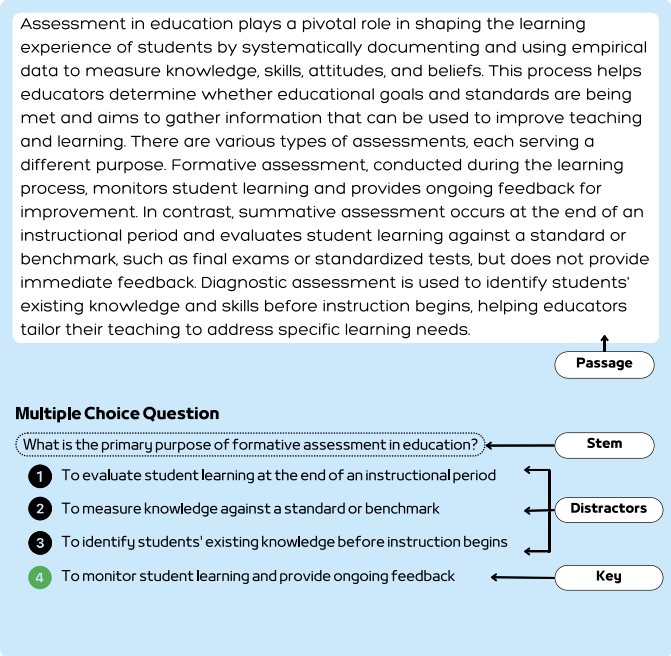


FIGURE 4. Component of MCQs: A detailed passage, stem, some distractors, and key.

TABLE 4. Common question formats in assessment.

Question Format	Description	Cognitive Level Focus
MCQs	Consist of a stem, correct answer(s), and distractors; widely used in various settings.	Different cognitive levels.
Short-Answer Questions (SAQs)	Require concise answers (1-3 sentences or 30-90 words) to demonstrate understanding; Very Short-Answer Questions (VSAQs) are a sub-type.	Factual recall, direct comprehension (SAQs); Low-level cognition (VSAQs).
Fill-in-the-Blank (Cloze)	Requires completing missing parts of a sentence.	Active recall, language skills, comprehension.
True/False Questions	Present a statement with two possible answers (true/false).	Basic knowledge, logical reasoning, factual or conceptual understanding.
Matching Questions	Use a two-column format to assess relational understanding and recognition; conceptually linked items are matched.	Relational understanding, recognition.
Essay Questions	Require in-depth responses to assess critical thinking, analysis, and synthesis skills; grading can be inconsistent [68].	Higher-order critical thinking.
Scenario-Based Questions	Present real-world scenarios requiring application of knowledge in practical contexts.	Application, analysis, decision-making, problem-solving.

The potential of LLMs extends beyond static QG. Current research is exploring innovative applications such as multimodal MCQs generated from diagrams and videos, adaptive learning systems that tailor question difficulty to individual student performance [48], and collaborative frameworks where AI-generated content is refined by human experts [16], [42], [50], [176]. These advancements promise to make educational assessments more personalized, engaging, and effective.

IV. FUNDAMENTALS OF MCQs

The MCQs format is a cornerstone of objective assessment due to its reliability and efficiency [18], [49], [174]. Its objectivity stems from a structure that minimizes scorer bias, ensuring consistent evaluation across a large number of respondents. A standard MCQs consists of three core components: a **stem**, which presents the question or problem; a **key**, which is the correct answer; and several **distractors**, which are plausible but incorrect options designed to challenge the respondent [31], [42], [69]. The effectiveness of MCQs hinges on the quality of its distractors. These are not merely incorrect answers, but they are carefully crafted to be attractive to respondents who lack complete understanding, often reflecting common misconceptions, partial knowledge, or logical fallacies. Thus, well-designed distractors are crucial for accurately discriminating between different levels of proficiency [69]. In reading comprehension tasks, a **passage** often provides the context from which the stem, key, and distractors are derived, with distractors representing

TABLE 5. Common types of MCQs.

Type of MCQ	Description
Single Correct Answer	One correct answer is provided among several options.
Multiple Correct Answers	More than one answer is correct, and the candidate must select all of them.
True/False MCQs	The candidate determines whether a given statement is true or false.
Matching Questions	The candidate matches items from two provided lists (e.g., terms and definitions).
Fill-in-the-Blank MCQs	The candidate selects the correct word or phrase from a list to complete a sentence.
Assertion-Reason MCQs	The candidate evaluates an assertion and a reason to determine if both are correct.
Sequence Ordering MCQs	The candidate selects the correct order or sequence of events from the given options.
Wh-questions	A question starting with a wh-word (e.g., who, what, where), with options.
Case-based/Passage-based MCQs	A series of questions are answered based on a provided passage or case study.

potential misinterpretations of the text, as illustrated in Figure 4.

While MCQs are one of several common assessment formats, as compared in Table 4, their structured nature makes them an ideal target for the automated generation techniques discussed in the following sections. The versatility of this structure also allows for various types of MCQs, each suited for different assessment goals, as shown in Table 5.

V. CLASSIFICATION OF METHODOLOGIES

The primary goal of automatic MCQs generation is to mimic the qualities of skillfully crafted manual questions, adhering to established psychometric principles of validity, reliability, and pedagogical utility [69]. A central challenge, highlighted by early research, is the creation of plausible yet incorrect distractors, a task that remains difficult for both manual and automated methods [31], [49]. To address the time-consuming nature of this process, researchers have developed a wide range of computational approaches.

Early methods like rule-based and template-based systems offered high control over question structure but lacked flexibility in handling the complexity of natural language [70], [71]. Subsequent data-driven approaches using statistical models, ML, and NLP improved adaptability by learning from text but often struggled with deep contextual understanding [3], [5]. More recently, the advent of DL models, and particularly LLMs like T5 and ChatGPT, has enabled significant improvement in generating high-quality, contextually relevant MCQs [7], [8], [13]. This section provides a structured classification of these methodologies, tracing their evolution and technical foundations. Each category of methodology is further summarized in a series of structured tables to facilitate comparison and analysis. This tabular format enables a concise yet comprehensive view of the technical approaches, performance measures, and constraints of individual studies, supporting a clearer understanding of methodological trends and research gaps across the literature.

A. FOUNDATIONAL APPROACHES

The earliest automated methods for MCQs generation were dominated by rule-based and template-based systems, which rely on explicit, handcrafted instructions to construct questions. These foundational approaches offered a high degree of control and predictability, making them effective for well-defined tasks.

Rule-based systems apply structured linguistic rules, pattern matching, and language-dependent criteria to generate question components from source text as given in Table 6. Common techniques included leveraging corpus word frequency and POS tagging for vocabulary cloze tests [65], using syntactic parsing to identify informative sentences in specific domains like sports [40], [130], and integrating semantic resources like a thesaurus with IRT for language proficiency exams [34]. Some systems advanced this paradigm by incorporating transformer models or SRL to improve distractor quality or outperform early neural methods in specific contexts [70], [76].

Template-based systems, a variation of this approach, use predefined sentence structures that are filled with specific content from a knowledge source as given in Table 7. This method is highly efficient for generating large volumes of structurally consistent questions. For example, a template-based system in medical education was able to generate 1,600 clinical reasoning MCQs in under two seconds [140]. More sophisticated implementations enhanced templates with advanced NLP techniques, such as semantic analysis to create more engaging questions for language learning [161], or integrated knowledge graphs and ML models to predict item difficulty [165].

Despite their utility, these foundational methods share significant limitations. Their reliance on manually crafted rules and templates makes them rigid, difficult to scale, and often domain-specific. Adapting them to new contexts requires substantial effort, and they frequently struggle to capture complex contextual nuances, leading to a limited diversity of generated questions that may require human correction to ensure quality [158].

B. THE ROLE OF NATURAL LANGUAGE PROCESSING

NLP provides the foundational toolkit for automatically generating MCQs from unstructured text [41], [43]. Systems leverage a broad spectrum of NLP techniques to deconstruct source material and build question components as given in Table 8.

Core methods include syntactic analysis, such as POS tagging and parsing, to extract key terms and understand sentence structure [82], [119]. Statistical techniques, like TF-IDF and statistical question ranking, are used to identify the most salient information within a text to form the basis of a question [135], [145]. More sophisticated approaches

TABLE 6. Rule-based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[34]	Thesaurus-based, Item Response Theory (IRT).	Travel	ATR's corpus	Human	Correlation with TOEIC scores: (80%)	6.5% of questions were flawed.
[40]	Topic modeling, parse similarity	Cricket	Wikipedia & news articles	Human	Sentence Selection: (93.21%), Key Selection: (83.03%), Distractor Quality: (91.07%)	Selects sentences with incomplete facts.
[65]	Word class tagging & frequency matching	General	Bank of English (BoE) corpus	Automatic	nth-word: (38-45%), frequency-based: (44-68%), word class-based: (50-58%)	Word class tagger has 5% error rate, ignores word sense & meaning.
[70]	Semantic Role Labeling (SRL)-based rules	Open-Domain	Custom corpus	Human	Usefulness rate: 71% (Yes/No) & 50% (wh-questions), outperforms neural baseline.	Struggles with complex sentences & subordinate clauses.
[76]	Dependency parses, RoBERTa ranker	Science	Wikipedia, SciQ	Accuracy	Improves accuracy from (63.4%) to (69.4%)	Relies on dependency parsing accuracy, requires distractor refinement
[104]	Pattern, Wikipedia entities	Cricket	Wikipedia	Human	Accuracy: Key Selection: (79.4%) Question Formation: (72.6%) Distractor Quality: (88%)	formation error: (27.4%), under- or over-informative: (21.2%)
[130]	Parse tree similarity, Named Entity Recognition (NER) filter.	Cricket	Wikipedia & news articles	Human	Sentence selection accuracy: (93.68%)	Requires a large set MCQs, rules were manually prepared.
[158]	Target word, lexicons	English	English-Chinese corpus	Human	System accuracy: (avg. >4.0/5.0) & convenience	Require human correction, lower satisfaction
[164]	Syntactic parse trees	Vocabulary	Custom corpus	Human	73% of generated items were rated as acceptable	Quality depends on large pool of syntactically similar phrases.
[168]	Part-of-speech (POS) tags, frequency & Dice coefficient	Open-Domain	Wikipedia	Human	Sentence selection accuracy: (92.37%) & answer-key identification: (95.90%)	Large input corpus required, incorrectly discards useful sentences.

TABLE 7. Template based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[140]	Cognitive model & item model	Medical	Turkish Report	Human	Acceptable: ((10/10)) & assessed clinical reasoning	Small evaluation scale, used static, non-generated distractors.
[161]	Event/entity co-reference & paraphrases	Biology	ProcessBank corpus	Human	Inference steps (0.76) with comparable grammar	Relies on expert-annotated text, low distractor quality.
[165]	Triple-pattern Knowledge Graphs (KG) queries	Open-Domain	Yago2s KG, Jeopardy	Hybrid	Difficulty classification accuracy: (66.4%), Kendall's Tau: (0.563) & Cohen's Kappa: (0.521)	Difficulty estimation accuracy is limited, requires an external annotated corpus.

focus on semantic understanding, using unsupervised relation extraction and dependency models to generate questions about the relationships between entities [106]. These diverse techniques have been adapted for numerous languages and domains, including Basque and Chinese, demonstrating the versatility of NLP as the backbone for MCQs generation [43], [131].

C. DATA-DRIVEN APPROACHES

The capabilities of NLP-based systems were significantly enhanced by the integration of ML and later DL models, which introduced data-driven learning to automate and improve the generation process. These approaches leverage algorithms to learn patterns directly from text, enabling more flexible and adaptable systems.

Early systems employed classic ML models and NLP techniques to automate different stages of the generation pipeline as given in Table 9. Supervised classifiers and Conditional Random Fields were used for tasks like identifying important sentences for question stems and generating cloze questions [79], [134], [151], [155]. Unsupervised methods, such as clustering, were applied

TABLE 8. NLP based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[106]	Semantic relations, dependency trees, distributional similarity.	Biomedical	GENIA, BioInfer	Human	Overall usability average (3.08/4.0)	Small-scale evaluation (two experts, 52 MCQs).
[131]	Linguistic rule transformations	Basque language	Basque corpus	Human	Acceptance rate for error correction (83.26%) & multiple choice (82.71%) questions.	Dependency on NLP tool quality, single expert teacher.
[135]	Inverse Document Frequency (IDF)	Physics	Wikipedia	Human	Acceptability rate: (70.66%)	Cannot handle mathematical formulae or symbols
[137]	Context Search, WordNet	English	NYTimes & WordNet	Human	Mistaken: (45%), Rated (3+ of 5) for quality: (59%)	Irrelevant or unnatural distractor quality.
[136]	WordNet, frequency matching	English	WordNet & BNC	Correlation scores	High correlation: (‘.61 < r < .85‘)	Limited coverage for words not in WordNet.
[77], [150]	Term extraction, Subject-Verb-Object (SVO) rules	Linguistics	Textbook, WordNet	Human & Psychometric analysis	Items rated "worthy": (57%), higher Discriminating Power	High reliance on human post-editing.
[74]	POS-based patterns	English	Wikipedia & VOA	Human	Acceptability: (77%), Error detection: (80%)	Relies on test patterns for each grammatical rule.
[138]	LexRank, SuperSense, POS	English	Transcripts & BNC corpora	Correlation & Human	Significant correlation: (‘r=0.57‘ & ‘r=0.35‘)	Moderate to low correlation
[143]	WordNet	English	WordNet	Informal testing	No quantitative results reported.	Lacks formal evaluation, low quality.
[144]	NLP tags, HowNet, & Word embeddings	Chinese	Wikipedia & HowNet	Human	Adequacy: (76%) for MCQs, (85%) for QA pairs, & 91% for questions.	Dependent on NLP tool accuracy, sentence segmentation
[145]	Distributional thesaurus, corpus	English	UKWaC corpus	Human	Acceptable: (53%), usable: (36%)	Carrier sentence quality issues.
[146]	POS tag counts, NER	General	Wikipedia & NER dataset	NA	No quantitative results reported.	Focuses on answer selection.
[119]	Term Frequency-Inverse Document Frequency (TF-IDF)	General	Lecture notes	Precision, Recall	Recall: (average >0.9), Precision: (average 0.36)	No evaluation of final MCQs quality.
[154]	T5 & Sense2Vec	General	Multiple datasets	Automatic	F1: (46.6%), BLEU-4: (19.59)	Lacks human evaluation and longer contexts.
[156]	Paraphrasing, Naive Bayes, Embeddings	General	SQUAD	NA	No quantitative results reported.	Lacks evaluation, very basic system
[157]	Link Grammar Parser	English	WordNet	NA	No quantitative results reported.	Need large knowledge base, lack evaluation
[84]	GloVe co-occurrence, N-grams matching	General	ReadWorks.org passages	Human	5-grams: (74%) fit narrow context vs. (13%) in full context	Poor filtering distractors, (12%) were acceptable.
[82]	SRL/NER tags	General	TREC QA	Automatic	N-gram similarity: 93.6% having at least one good distractor.	No human evaluation, tools dependent
[171]	Weighted features, similarity	Biology	Biology textbook	Human	Sentence selection: >79% & key selection: >84%,	Relies only on source text, lack of semantic features

to create plausible distractors from a given text [45]. While these methods proved effective for specific tasks and languages, including Chinese and European Portuguese, they were often limited by the need for extensive feature engineering, their reliance on external knowledge bases like WordNet, and the scarcity of high-quality training data [42], [75].

After the emergence of DL models, natural language developed a new level of refinement. One of the core algorithmic program used by these models is their advanced neural architectures to develop the capability of capturing sequential patterns. Some of the most studied models like RNNs, LSTM, and encoder-decoder could capture complex, long-range dependencies in language, leading to the generation of more fluent and contextually coherent questions as given in Table 10 [39], [42], [147]. This has made it possible for the development of systems in specialized domains such as medicine and pharmacy, where cognitive models were integrated to ensure psychometric validity [142]. These modern systems continue to excel in a broader spectrum of NLP tasks, including POS tagging, NER parsing, and semantic similarity measures. These techniques are mainly used to extract key information, structure questions, and generate plausible distractors [82], [106], [119], [135]. The automation and quality of MCQs have significantly improved by these data-driven advancements, though challenges related to dataset dependency and the need for expert validation persist.

TABLE 9. ML based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[45]	Keyword extraction, K-means cluster	Indian History	Web documents	Human	Stem Accuracy: (>93%), distractor accuracy: (71.86%)	Initial keyword dependent, non-standard evaluation.
[57]	Naive Bayes, Support Vector Machine (SVM)	Technical	Textbook	Human	Quality: (74.38%) & Grammatically correct: (97.24%)	Time-consuming, limited to question stems and keys.
[56], [78]	Naive Bayes, K-Nearest Neighbors	English	BBC articles	Human	97.6% of blank positions were rated as possible	Insufficient features, unevaluated random DG.
[75]	Regression model, mixed features	Chinese	Textbooks & HowNet	Psychometric analysis	ID: (0.41), Distractor Usefulness: (0.82)	Incomplete character coverage in HowNet
[79], [134]	Preference Learning & statistical patterns	English	TOEIC workbooks	human	Highly ranked: (91.83%), recall: (18.91%)	Low recall, struggles with wider context
[120]	NLP heuristics & parsing	General	Wikipedia articles	Human, ROC curves	True Positive: (83%), False Positive: (19%)	No distractors generation, only gap selection
[151]	SVM, feature engineering	Language	CETEM-Público	Precision, Recall, & F-Measure	F-Measure: (75.5%), reduced false positives.	Stems rejected: (87%), model/expert mismatch
[155]	Topic modeling & parse similarity	Sports	Wikipedia & news articles	Human	Sentence: (93.21%), Key: (83.03%), Distractor: (91.07%)	Selects sentences with incomplete facts.
[159]	Discriminative models (SVM)	English	Lang-8 Corpus	Human	Distractor reliability (RAD): (98.3%), correlation: (0.76)	Performance impacted by parsing noise in corpus.

TABLE 10. DL based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[142]	Cognitive model, item model	Pharmacy	PEBC, NAPRA, textbooks	Human	Generated 15,100 MCQs, satisfied all 8 item-quality guidelines	Focused on nausea/vomiting, no psychometrics yet.
[39]	Cognitive model, item model	Medical	Human	NA	Generated 1248 unique items from one model in 6 hours	Not validated or generalized, no evaluations.
[147]	Custom model	Education	RACE	Human, Automatic	Questions BLEU-4: (11.64) Distractors BLEU-4: 4.45	Performance gap, auto metrics ill-suited for distractors.
[152]	T5, sense2vec	Education	SQUAD1.1, RACE	BLEU, F1, Exact Match	T5 on SQUAD: 63.28 F1, 37.24 Exact Match	No collaborative platform, no answer justifications.
[17]	T5, LeNet, TextTiling	Learning	Custom dataset	Accuracy	Avg. accuracy for image linking: (70%)	Poor transcription & language, unusable questions.
[98]	doc2vec, SBERT	Law	Custom corpora	Cosine similarity, Human	Usable QA pairs: (57%)	Limited corpus size with old data

D. KNOWLEDGE-BASED APPROACHES

Data-driven methods totally depend on the quality and availability of data. Unlike data-driven methods that learn from unstructured text, ontology-based systems leverage formal, structured knowledge representations to generate questions that are conceptually coherent and logically sound. Most of these systems take advantage of domain ontologies, as they are often built with Web Ontology Language (OWL), to model the concepts and relationships within a specific subject area.

This approach enables for a high degree of control over the semantic content of the questions. For instance, the system developed by a group of researcher known as "OntoQue" used semantic web technologies to generate high-quality MCQs, with 90% of its output rated as "good" by evaluators [35], [80]. Ontologies have been successfully applied in specialized domains like medicine for generating case-based questions [81], and have been used to create specific question types, such as those based on analogies or concept similarity [36], [38]. They have also been shown to improve the quality of distractors by providing semantically related but incorrect concepts from the knowledge structure [160]. However, the primary limitation of this paradigm is its heavy reliance on the existence of high-quality, comprehensive ontologies, which are expensive and labor-intensive to create. Furthermore, while conceptually robust, these systems often struggle to generate syntactically varied and linguistically complex questions without sophisticated natural language generation components [132], [163].

TABLE 11. Ontology based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[81]	Ontology-based System (EMCQG)	Medical	EMMET-OWL	Human	Appropriate: (79%)	Ontology lacks specific context, linguistic issues.
[35], [80]	Ontology-based engine (OntoQue)	History	HistOnto, SemQ	Precision, Human	Avg. precision ('Good' items): (90%)	Knowledge/recall level questions.
[73]	SimpleNLG	Academia	Eupalineio Tunnel	Human	Rated 'satisfactory for assessment' by experts	Lack syntactic correctness & cognitive skills.
[132], [163]	Node & edge label-set approaches	General	Harry-Potter, Geographical ontologies	Human	Difficulty hardness measure: 65% of option sets	Basic syntactic generation, ignores stem complexity.
[36]	Ontology-based analogy generator	General	Pets, Pizza ontologies	Solver accuracy	88% solver accuracy on Pizza Ontology questions	Low solver accuracy, only class-subclass relations.
[133]	Pattern-based QG (SAVANT)	Medical	SNOMED-CT, ICD9 ontologies	Automatic	80% of mappings validated or corrected	Cannot handle contextual answers.
[37], [38]	Similarity measures	Education	Custom ontology	Human	Usefulness: (63/65); Difficulty prediction: (51/65 agreed with $\geq$1 reviewer)	Costly, Grammatical errors.
[160]	Ontology graph-structure	Biology	Biology ontology	Human	Positive ratings: 66% (60/90)	Relies on correctness & completeness of ontology

TABLE 12. LLMs based MCQs generation systems using fine-tuned models.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[7]	T5-based pipeline	General	SQuAD, DG-RACE	Automatic & Human	Question quality: (4.28/5.0), BLEU-1:14.80, ROUGE-L: 14.91, Cosine Similarity: 0.71	Test retention more than comprehension, automatic DG metrics are inadequate.
[86]	Fine-tuned GPT-3	General	EQG-RACE & RACE	Automatic & Human	Quality score: (90.5%)	Metrics are problematic, low distractors quality.
[97]	mT5-based pipeline	Spanish	SQuAD, SciQ, CosmosQA	Automatic	ROUGE-L: (29.86)	Machine-translated data & multilingual model.
[11]	LLMs (Llama 2, GPT-4)	Radiology	ABR blueprint, ACR DXIT	Human, Accuracy	100% answer accuracy	Text-only questions, basic evaluation, single prompt strategy used.
[15]	OpenBioLLM-70B, GPT-4o, Claude 3.5	Medical	Semantic Scholar, DailyMed	Human	GPT-4o outperformed, quality metrics and topic diversity: (89.8%)	Requires final human review, text-only questions, evaluated on one specialty.
[89]	KB/BERT, GPT-3, SweCTRL-Mini,	Swedish	SweQUAD-MC, Quasi, Plugga	Human	ChatGPT 'acceptable' MCQs: (63.7%), outperforming other models	Small fine-tuning datasets, may lack pedagogical quality.
[92]	Self-refinement framework	Medical	MedQA, USMLE	Human, Cohen's kappa	72.5% win rate over baseline GPT-4	Limited generalizability, LLMs-as-Judge has self-evaluation & position bias.

E. LLM-BASED APPROACHES

The advancement in the field of generative AI and the rapid development in the research of LLMs represents the current state-of-the-art in automated MCQs generation. Currently, researchers are making use of their advanced capabilities in comprehension, summarization, logical reasoning, instruction following and multilingual text generation [67], [186]. As shown in Figure 5, models like GPT-4 and Gemini can produce high-quality, contextually relevant questions that align directly with specific learning objectives [7], [18], [54], [58], [172], [180]. Researchers have developed several distinct strategies for leveraging these models to their full potential in the respective domain, which are classified in the following sections.

1) FINE-TUNING LANGUAGE MODELS

Fine-tuning adapts a pre-trained LLMs for a specialized task, such as question or DG, with several studies detailed in Table 12. This approach is common for models like T5, which has been fine-tuned on datasets like DG-RACE to improve distractor quality, and for BERT-based encoders used for question formulation [7], [89]. In specialized domains like

TABLE 13. LLMs based MCQs generation systems using prompts.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[139]	Prompt-based GPT-3.5	Medical	NA	Automatic	ID: (0.41),correlation: (0.39)	lacked qualitative analysis.
[141]	Prompt-based GPT-3.5	Medical	N/A	N/A	Presents a prompt to generate plausible, well-constructed MCQs in seconds	No empirical evaluation, questions quality unverified.
[85]	GPT-4 pipeline	Software Engineering	Lecture materials	Human	All 6 experts agreed the tool saves time in quiz generation	Lack versatility, erroneous distractors, small expert evaluation (n=6)
[99]	GPT-4, JSLM-7B	Medical	Nursing Exams	Automatic, Human	Distractor variation (DV): 0.126,	Failed to capture creativity, performance disparity between model sizes.
[16]	GPT-4 & Human-LLM tool (HEDGE)	Maths	Math Data	Human	Validity: (70%), distractors validity: (37%)	Limited in generating distractors based on common student errors.
[19]	Prompt-based LLMs (GPT-3, GPT-4)	Computing	Canterbury Question Bank	Accuracy, correlation	Correct answer for 78.5% of MCQs	Did not evaluate distractor quality, limited prompt engineering.
[149]	Prompt engineering (GPT-3.5, GPT-4)	STEM	SME-developed Q&A pairs	Human, Cohen's kappa	Answer accuracy: (1.85/2.0)	Limited to 3 STEM subjects.
[91]	Customized ChatGPT, RAG	Medical	Past Exams	Human, Accuracy	Answer accuracy: (100%), quality ratings: (6.7-9.8/10)	Subjective evaluation, not tested on students.
[95]	Multi-Stage Prompting, GPT models	Multilingual	SQuAD, GermanQuAD, HiQuAD	Automatic, Human	Outperforms single-stage baseline in automatic and human evaluations	Lower performance on low-resource languages,limited human evaluation.

TABLE 14. LLMs based MCQs generation systems using pipeline and modular approaches.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[47]	GPT-4, RAG, self-refine	Computer Science	Custom Data & MCQs	Human & LLMs expert	High expert ratings for Grammatical Fluidity and Relevance	Small sample size, limited to one domain (Computer).
[96]	LLMs (GPT-4-Turbo, Llama-2)	Education	Turkish-Quiz-Instruct	Human & Automated	GPT-4-Turbo: 85% of MCQs rated 'useful' by humans	Generates factually incorrect short-answer questions.
[88]	Personalized Cloze Test Generation (PCGL) Framework	Education	CLOTH	GPT-4, difficulty adjustment	Outperformed baseline in difficulty-adjusted sentence & DG	Room for improvement, some experimental results not ideal.
[175]	GPT-3.5, RoBERTa	Chemistry, Biology	OpenStax textbooks	Automatic, Human	80% machine alignment for "Evaluation" level	Misalignment, ambiguity and lower quality
[90]	Llama 3 & GPT-variant	General	Custom resolutions	Human, Automatic	94.4% valid questions	Non multi-hop questions, LLMs shared biases.
[94]	T5, TextTiling, LeNet	E-learning	Custom video lectures	Accuracy, Human	Avg. accuracy: (70%)	Poor transcription and conversational language.
[20]	Systematic method, GPT-4o	Materials Science	Scholars Databases	Accuracy, Human	7k dataset for benchmarking LLMs, Accuracy scores: (0.76 - 0.777)	Performance varies across LLMs, safety judgment remain challenging.

medicine, this technique allows cost-effective Bio-LLMs to achieve high-quality results in complex pipelines [11], [15]. Despite its power, fine-tuning does not inherently solve persistent issues like knowledge cutoffs, factual hallucinations, or limited ID, which require ongoing research into better evaluation frameworks [92].

2) PROMPTING LLMs

As an alternative to fine-tuning, prompting leverages the in-context learning capabilities of large-scale models like GPT-4 through carefully engineered instructions, as shown in the Table 13. Recent work has moved beyond simple zero-shot prompts to more sophisticated strategies. Techniques like RAG ground the model in external knowledge to improve factual accuracy, while Chain-of-Thought (CoT) prompting and self-refinement encourage more logical reasoning, leading to better adaptive quizzes [47]. Multi-stage prompting has also been shown to improve distractor quality across multiple languages [95], while prompting accelerates development in fields like computing, education and nursing [85], [99]. Studies demonstrate that performance varies by model and prompt design, and human-in-the-loop

TABLE 15. LLMs based MCQs generation systems using multi-agents.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[53]	Langchain, GPT-4o	Mathematics	NA	Human	Over 90% of generated questions rated 'useful' by experts	Specific to maths.
[12]	Multi-agent system (UsmleGPT), Prompting	Medical	NBME framework	Human	Generates USMLE Step 1 items through a simulated NBME workflow	Potential for systematic biases in LLMs, cost considerations.
[184]	Langchain, GPT4,	AI Literacy	Learning objectives	Human	84.2% of questions rated as suitable for classroom use by experts	Rephrasing issues, Bloom's level and grade level appropriateness.
[183]	AutoConverter	Model Evaluation	VMCBench	Human	Vision models showed similar or lower accuracy on AutoConverter questions	Relies on the quality of initial open-ended VQA datasets.
[182]	Langchain, Gemini AI	Education	User files	Human	Automated MCQs and evaluation from various file formats	Quality depends on input, English only

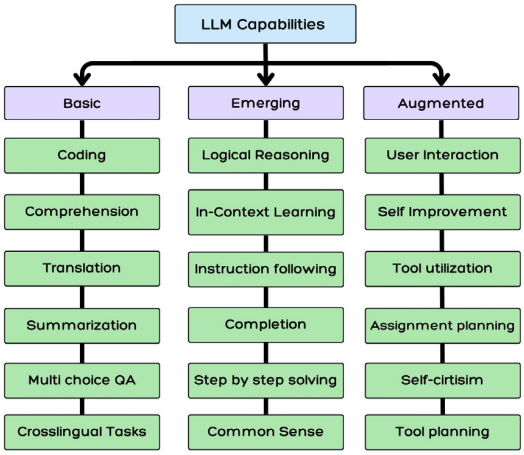


FIGURE 5. LLMs capabilities. Courtesy by [180].

validation remains essential to address domain-specific errors and generate challenging, error-based distractors [16], [19], [139], [141], [149].

3) PIPELINE AND MODULAR APPROACHES

Many systems, such as those listed in Table 14, adopt a pipeline architecture, breaking the complex task of MCQs generation into sequential stages. This modularity allows for the integration of different models or techniques at each step. For example, a pipeline might use a RoBERTa model to enforce alignment with Bloom's Taxonomy before passing the output to GPT-3.5 for generation [175]. Such architectures have been applied in diverse domains, from generating Turkish quizzes to legal and materials science questions [20], [90], [96]. While these systems can produce highly contextualized questions, especially when enhanced with RAG and CoT reasoning [47], they underscore the need for better dynamic difficulty calibration and more robust evaluation metrics that capture cognitive depth [94].

4) MULTI-AGENT SYSTEM APPROACHES

A more recent innovation involves multi-agent systems, where different LLMs agents collaborate in roles such as "generator," "critic," and "evaluator" to refine the final output as given in Table 15. This approach simulates an expert human workflow and is particularly useful for high-stakes assessments. For instance, systems combining models like GPT, Claude, and Llama with CoT-SC and Tree-of-Thought (TOT) mechanisms have been used to emulate the National Board of Medical Examiners (NBME) item-development process [12]. Similarly, LangChain-powered agents have been orchestrated to generate contextualized mathematics questions that outperform single-model prompts [53] and to automate end-to-end quiz creation using Gemini [182]. While promising, these systems face challenges related to computational cost, potential for bias, and the need for real-world validation [183], [184].

5) HYBRID LARGE LANGUAGE MODELS APPROACHES

Hybrid methods, showcased in Table 16, combine the generative power of LLMs with other techniques to enhance accuracy and contextual relevance. This often involves integrating LLMs with rule-based components, lexical databases like WordNet, or structured knowledge from ontologies. For example, a pipeline might combine T5 with traditional NLP methods like POS tagging to produce better semantic distractors [102]. In other cases, RAG systems are compared with fine-tuned models like Llama to determine the optimal approach for a given task, revealing that specialized fine-tuning can yield superior distractor quality [46]. Multi-task learning with models such as BART and ProphetNet has also been shown to boost student engagement by jointly training on generation and evaluation objectives [148]. These hybrid strategies highlight a pragmatic trend of combining the "best of all worlds" to overcome the limitations of any single approach [14], [153].

F. HYBRID APPROACHES FOR MCQs GENERATION

Hybrid approaches combine ideas from different methodologies in an effort to make the best use of what each does well while compensating for their individual shortcomings. Table 17 gives an overview of some representative systems. The idea itself is simple but effective: instead of relying on one technique, these systems bring together complementary

TABLE 16. LLMs based MCQs generation systems using hybrid approaches.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[102]	T5, sense2vec, POS Tagger	Education	SQUAD, CBSE Textbooks	F1, Human	Fluency: (5/5), grammar (5/5), answerability (4/5), relevance (4/5), F1: 48.58	Not able to handle answer phrases of varying length.
[14]	Macaw+, Bing Chat	Education	Science textbook	Human	LLMs performance similar to human on 6 of 7 metrics	Structure issues in answers and options
[148]	Multi-task approaches (T5, BART, ProphetNet)	Education	Custom & SQUAD	Automatic, Human	T5 performed best, QA pairs improved student performance.	Limited to computer science, only evaluated three LLMs.
[46]	LLMs-based agents, RAG, Fine-Tuned LLMs)	Education	Custom courses	Human, Mixed-effects modeling	Provides a framework for aligning AI-generated content with learning objectives	Bias and contextual limitations.
[153]	GPT-3.5-based pipeline (VocaTT)	English	Academic word list	Human	75% well-formed sentences, 66.85% suitable word options	Needs improved POS tagging, sentence validation, and prompts.

TABLE 17. Hybrid based MCQs generation systems.

Ref.	Method / Model	Domain	Dataset(s)	Evaluation Metrics	Key Results	Limitations
[83]	TF-IDF, Jaccard similarity, K-means, BERT	Computer Science	Books	Human	Overall MCQs accuracy: (80%)	Need further improvement, domains specific
[44]	WordNet, neural embeddings & Jaccard similarity	Education	NCERT textbooks	Human & Automatic	'worthy': (95.1%), 'utilizable': (94.5%), distractor closeness: (99.07%)	Simple sentences, no multi-line facts or non-text content.
[93]	Zero-shot pipeline, T5 & BERT	Skill Assessment	CoNLL-2003 & Quora datasets	Human & Automatic	Semantic similarity (4.33/5) & fluency (4.34/5), precision (97.7%)	Application specific, low precision for questions
[100], [162]	Topic Models, SVM Classifier, word2vec	Education	Wikipedia, Amazon Mechanical Turk	Human & Automatic	Gap selection accuracy: (81.0%), distractors as good/fair: (94%)	Sentences aren't independent, Low inter-annotator for distractors
[105]	Dynamic Ontology (QG-SKI), TF-IDF	E-Learning	LOD Cloud, Wikipedia	Accuracy	Accuracy: (98.15%), outperforms baseline models	Not specified
[87]	POS tagging, Linear Regression, WordNet	Education	ORCHID corpus, WordNet	Human	Acceptability: (81.32%), ≥1 acceptable distractor: (49.32%)	Low distractor quality: (49.32%), errors in processing
[170]	Sentence splitting, Parsing	English	Project Gutenberg	Human	Overall acceptability: (53%), answer accuracy: (70%), passage accuracy: (97%)	Acceptability (53%) is low, high error rate in correct answer
[169]	Production rules, LSTM Model, NLP	Education	Custom quiz & Wikipedia	Human	Difficult to differentiate between automatic & manual questions	Difficulty in connecting sentences, technical distractor, "Wh"-question classification
[103]	Preference learning, Hypernym/Hyponym similarity	E-assessment	Web corpus, Wikipedia	Accuracy	Summariser accuracy: (72%), blank accuracy: (77.6%), distractor accuracy: (78.8%)	No pronouns, no co-reference resolution, limited analogy questions
[50]	NLP, BART, PEGASUS	Education	CNN-DailyMail, Wiki-FactCheck	Human	High usability & satisfaction, adoption: (59%), strong preference	Small sample size, no student evaluation, difficult DG.
[51]	Template-based, Custom GPT	Medical	GPT Builder	Time	Time reduction: (5 hours to <10 mins), AI generated plausible models	Proof-of-concept, no empirical validation, simple templates only

tools to get more reliable and flexible results. For example, statistical models are often used for fast keyword extraction, NLP methods help maintain grammatical accuracy, and deep learning (DL) models add a layer of semantic understanding that traditional methods lack.

In recent work, researchers have often paired classical statistical techniques such as TF-IDF with newer neural models like BERT to improve question generation in specific areas, including computer science [83], [105]. Others have taken a more layered approach by linking a sequence of NLP tasks, such as entity recognition, parsing, semantic analysis with neural components to produce questions that read more naturally and make better sense contextually [44]. A slightly different but related line of work focuses on human–AI collaboration, where automated tools handle the heavy lifting and educators refine the output. This approach,

TABLE 18. Comparison of MCQs generation methodologies.

Methodology	Core Idea	Requires	Flexibility	Control	Strength	Weakness	Output
Rule-Based	Explicit rules	Hand-made rules	Low	High	Predictable	Brittle, scale	Follows rules
Template-Based	Fill patterns	Templates	Low	High (structure)	Speed, consistent	Repetitive	Consistent
Statistical/ML	Find data patterns	Training data	Moderate	Low	Hidden patterns	Needs data	Reflects data
Ontology-Based	Structured knowledge	Knowledge maps	Moderate	Medium	Conceptual accuracy	Needs good maps	Logically sound
Deep Learning	Learn complex patterns	Large datasets	High	Low	Context, fluency	Needs data, errors	Fluent (mostly)
NLP Techniques	Specific language tools	Text, tools	Moderate	Medium	Specific tasks	Complex pipelines	Varies by tools
LLMs	Massive models	Huge data, compute	High	Medium	Fluency, context	Mistakes, check needed	Very fluent, needs check
Hybrid	Combine approaches	Multiple components	Varies	Varies	Best of worlds	Complex build	Varies

TABLE 19. Comparison of LLMs-based vs. Classical methods for MCQs generation.

Aspect	Classical Methods	LLM-Based Methods
Fluency	Rigid, template-bound output [65], [70].	Fluent, human-like sentence variety [7], [18].
Context	Limited to predefined patterns [158], [160].	Strong contextual awareness [47], [149].
Distractors	Semantically grounded via ontologies/WordNet [36], [38], [137].	Often superficial, word-level associations [16], [19].
Accuracy	High from verified sources [81], [133].	Prone to hallucinations [10], [141].
Scalability	Low, manual rule/ontology creation needed [71], [77].	High; adapts via prompting [95], [96].
Adaptation	Time-intensive new rule sets required [81], [140].	Prompt engineering or fine-tuning [15], [47].
Cognitive Control	Explicit via Bloom's-aligned templates [1], [69].	Inconsistent level targeting [175], [186].
Oversight	Moderate; predictable, systematic errors [158].	High; unpredictable errors [49], [181].
Explainability	Transparent, traceable process [35], [80].	Opaque generation.
Speed	Fast after setup; setup is slow [140].	Fast with minimal setup [51], [91].

which has gained traction in domains like medicine, helps maintain quality while still saving time [50], [51]. Even with these advances, hybrid systems are not without problems. Models tuned for one subject or language often fail to generalize, and distractor quality can vary widely across new contexts [87], [100].

To offer a broader perspective, Table 18 summarizes key characteristics, advantages, and weaknesses of the main methodological families. Rule-based and template-driven systems allow close control over structure but tend to be rigid. Statistical and machine learning (ML) approaches detect meaningful patterns but may overlook contextual nuance. Ontology-based systems rely on structured domain knowledge and thus provide conceptual coherence, though they are limited by the completeness of available ontologies. More recently, methods rooted in NLP, deep learning, and LLMs have achieved impressive results in generating fluent and contextually appropriate questions. Yet even these sophisticated systems still face well-known challenges, such as maintaining factual accuracy, producing balanced distractors, and incorporating sufficient human oversight.

Hybrid frameworks can be seen as a natural response to these trade-offs. By combining complementary strengths, they aim to create systems that are not only technically robust but also educationally meaningful. From a broader perspective, each methodological tradition, whether rule-based, statistical, or neural, contributes valuable insights to this evolving area of research. The gradual blending of these ideas shows both how far the field has come and how much room there still is for improvement.

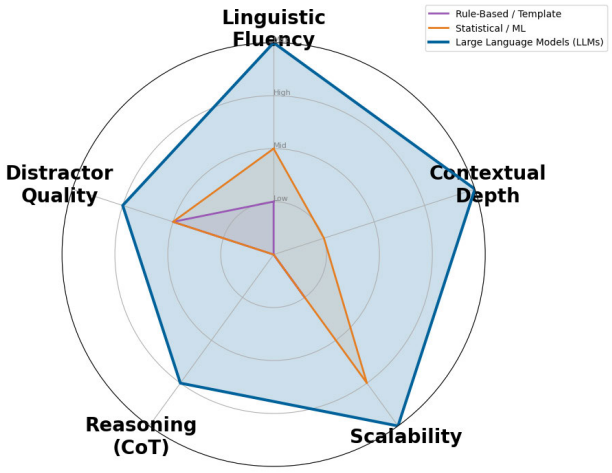


FIGURE 6. Radar chart comparing MCQs generation methodologies across five key dimensions.

To provide additional clarity regarding the trade-offs between traditional and modern approaches, Figure 6 provides a visual summary of how each methodology family performs across five key dimensions. LLMs-based

TABLE 20. Popular datasets used in MCQs generation. This table lists datasets with features including: Domain (topic area), Q (number or type of questions), Q-Type (format of questions), A-Type (answer structure), Q-Source (origin of questions), A-Source (source of answers), MR (machine reading comprehension support), and CS (commonsense reasoning requirements).

Dataset	Domain	Q	Q-Type	A-Type	Q-Source	A-Source	MR	CS
BoolQ [22]	Wikipedia	16k	Yes/No	Boolean	Search Query	Wikipedia	Yes	Yes
CLOTH [113]	English Exams	99k	Cloze	MCQs	Teachers	Exam passages	Yes	Yes
CELA [112]	English Cloze	3k	Cloze	Binary, 3-class	Human, Automatic	English Exams	No	Yes
CQA [55]	Common sense	12k	MCQs	Single word	Crowd workers	conceptnet	Yes	Yes
COSMOS QA [26]	Common sense	35k	MCQs	Single word	Crowd workers	Webblogs	Yes	Yes
DGen [114]	Open-domain	2.9k	MCQs	Single word	-	Multi-Sources	Yes	Yes
DREAM [109]	Dialogue	10k	MCQs	Free-form text	Teachers	Dialogues	Yes	Yes
EduQG [24]	Education	3.4k	MCQ, Cloze	Single word	Educators	Textbook	Yes	Yes
HEAD-QA [128]	Healthcare	6.5k	MCQs	Single word	-	Publications	Yes	No
HotpotQA [64]	Wikipedia	113k	Multi-hop	Single word/list	Crowd workers	Wikipedia	Yes	Yes
LearningQ [32]	Education	230k	MCQs	Single word	Instructors	Online Courses	Yes	Yes
MCTest [28]	Fictional stories	2k	Comprehension	MCQs	Crowd workers	Fictional stories	Yes	-
MedMCQA [23]	Medical	194k	MCQs	Word or list	PG Exams	-	Yes	Yes
MovieQA [116]	Movie	15k	MCQs	Single word	Human	Scripts	Yes	Yes
MS MARCO [63]	Web	1M	Factoid	Human-generated	Search Query	Bing Search	Yes	Yes
MythQA [59]	Twitter	522	Open-domain	Multiple, Tweets	Human annotators	Twitter	Yes	Yes
NarrativeQA [129]	Stories	47k	QA	Free-form text	Crowd workers	Scripts	Yes	-
NQs [61]	Wikipedia	307k	QA	Long/Short	Search Queries	Wikipedia	Yes	Yes
NewsQA [126]	News	100k	Factoid	Text span	Crowd workers	CNN	Yes	No
NewsQuizQA [33]	News	20k	MCQs	Single word	Human writers	News Summaries	Yes	Yes
OpenBookQA [127]	Textbook	10k	Open-book	MCQs	Textbook	Textbook	Yes	Yes
QuAIL [108]	Multi-domain	15k	Comprehension	Multiple	Expert writers	News & Articles	Yes	Yes
QUASAR [27]	Open-domain	80k	Cloze & Trivia	Text span	Internet	Internet	Yes	Yes
Quasi [25]	Swedish	436	Comprehension	Multiple	GPT-3	Swedish Exam	Yes	Yes
QASC [110]	Science	10k	Multi-hop	Multiple	Crowd workers	Text Corpus	Yes	Yes
RACE [30]	English	100k	MCQs	Single word	Instructors	Written text	Yes	Yes
ReClor [107]	Reasoning	6k	MCQs	Single word	Exams	-	Yes	Yes
ReCoRD [122]	News	121k	Comprehension	Cloze	News articles	News articles	Yes	Yes
RecipeQA [123]	Recipes	37k	Multi-modal	MCQs	Recipes	Recipes	Yes	Yes
ROCStories	Common sense	50k	Stories	Sentence	Crowd workers	-	Yes	Yes
SCDE [111]	Sentence Cloze	30k	Sentence Cloze	Candidates	English Teachers	Exam passages	Yes	Yes
SciQ [29]	Science	14k	MCQs	Single word	Crowd workers	Science textbooks	Yes	Yes
SQuAD [60]	Wikipedia	108k	Factoid	Text span	Crowd workers	Wikipedia	Yes	No
SocialQA [124]	Social	38k	QA	Multiple	Crowd workers	Social events	Yes	Yes
SQs [173]	Freebase	108k	Factoid	Object	Freebase KB	Freebase KB	No	No
SWAG [121]	Common sense	113k	MCQs	Sentence	-	Video captions	Yes	Yes
TriviaQA [62]	Open-domain	95k	QA	Free-form text	Trivia fans	Web	Yes	Yes
TQA [117]	Textbook	26k	Multi-modal	MCQs	Textbook	Textbook	Yes	Yes
TabMCQs [21]	Science	9k	MCQs	Single word	Crowd workers	Knowledge Tables	Yes	Yes
WikiQA [127]	Wikipedia	3.8k	QA	Free-form text	Wiki	Wikipedia	Yes	Yes

approaches excel in linguistic fluency, contextual understanding, and scalability, while classical methods show relatively stronger performance in distractor quality. Table 19 presents a more detailed side-by-side comparison across ten aspects. Classical methods including rule-based, template-driven, and ontology-based systems offer predictable outputs and strong control over question structure, but they require substantial upfront effort and do not adapt easily to new domains. On the other hand, LLMs-based methods excel in fluency, scalability, and contextual understanding, yet they come with their own set of challenges. For example, factual accuracy is harder to guarantee, distractors may lack pedagogical grounding, and the generation process is largely opaque. Neither family of methods is strictly better as the choice depends on the specific requirements of the assessment task at hand.

VI. DATASETS

The development of robust automatic MCQs generation systems relies on a wide range of publicly available datasets for training, fine-tuning, and evaluation. Table 20 provides a comprehensive summary of key datasets used in the field, detailing their domain, size, question-and-answer formats, and the cognitive capabilities they are designed to test, such as machine reading comprehension (MR) and commonsense (CS) reasoning.

From a technical standpoint, these datasets exhibit significant diversity in their structure and complexity, which directly influences the design of generation models. Many datasets, such as SQuAD [60], NQs [61], and TriviaQA [62], originate from question-answering (QA) tasks and thus require models to first formulate a question-answer pair before transforming it into an MCQs format. In contrast, datasets like CLOTH [113] and RACE [30] provide a more direct training setup, with native cloze and multiple-choice formats, respectively. The variety in answer types ranging from single words and text spans to free-form sentences also introduces different levels of complexity, particularly for the challenging task of DG. Furthermore, the scale and nature of these datasets present distinct opportunities. The large size of datasets like MS MARCO [63] and LearningQ [32] enables the training of robust DL models, while the inclusion of complex reasoning requirements in datasets such as HotpotQA [64] and OpenBookQA [127] drives the development

of models with advanced inference capabilities. The diverse sources, from Wikipedia and textbooks to crowd-sourced annotations, introduce potential biases and stylistic variations that models must learn to navigate.

VII. GENERIC ARCHITECTURE

The design of most automatic MCQs generation systems is inspired by the manual authoring process, following a multi-phase pipeline to transform unstructured text into a valid assessment item. While specific methodologies vary, a generic workflow can be abstracted into three core phases: content identification, item construction, and quality assurance, as illustrated in Figure 7.

A. CONTENT IDENTIFICATION AND SELECTION PHASE

The initial phase focuses on preparing the source text and identifying the core information that will form the basis of a question. This involves three main steps:

1) PRE-PROCESSING

The raw input text is cleaned, normalized, and structured. This includes tasks such as sentence simplification [65], [78], [106], [150], lexical analysis, and syntactic analysis using POS tagging and parsing [34], [65], [77], [131], [150], [171], [177]. Ambiguities are often resolved through techniques like co-reference resolution and Word Sense Disambiguation (WSD) [77], [150], [161].

2) SENTENCE SELECTION

From the processed text, candidate sentences are selected based on their suitability for generating a question. Criteria often include sentence length, syntactic simplicity, and the presence of strong grammatical structures like SVO patterns [34], [40], [151], [171], [177]. ML and DL models are frequently used to classify and rank sentences based on these features [73], [78], [79], [134], [151].

3) KEY SELECTION

Within a chosen sentence, the ''key'' is identified. This is typically a key term, noun phrase, or verb that represents a central idea. Common selection techniques include frequency-based metrics like TF-IDF [65], [77], [150], [171], syntactic cues from POS and parse information [34], [72], [131], [150], and ML models trained to predict the most likely answer phrase [78], [79], [134].

B. ITEM CONSTRUCTION PHASE

Once the core content is identified, the system constructs the components of the MCQs.

1) QUESTION GENERATION

The selected sentence is transformed into a question stem. This is commonly done by converting the declarative sentence into an interrogative one, often by replacing the key with an appropriate wh-word (e.g., ''who,'' ''what,'' ''where'') based on its semantic type [40]. Syntactic transformations using dependency patterns or SVO structure are also employed to rephrase the sentence into a natural-sounding question [77], [106].

2) DISTRACTOR GENERATION

This critical step involves creating plausible yet incorrect answer choices. As summarized in Table 21, a variety of techniques are used, including leveraging lexical resources like WordNet to find synonyms [34], [77], [79], [134], [150], using domain ontologies for conceptually related terms [73], applying the distributional hypothesis to find words that appear in similar contexts [72], [106], and ensuring grammatical consistency with the key using POS information [65], [77], [79], [134].

C. QUALITY ASSURANCE AND REFINEMENT PHASE

The final phase focuses on ensuring that the generated MCQs are not only grammatically sound but also accurate, relevant, and suitable for practical use in an educational environment.

1) VERIFICATION

At this stage, each generated question goes through a series of automated and, at times, semi-automated quality checks. These typically include grammar and syntax validation [150], along with methods that assess semantic consistency, such as NER and semantic role labeling [2], [65], [77], [131], [150]. To evaluate the plausibility of distractors, similarity metrics are commonly applied, while predictive models may help estimate how difficult each question is likely to be for learners [93], [170]. Although these automated tools are useful, their results often require manual confirmation to ensure that subtle conceptual errors are not overlooked.

2) POST-PROCESSING

Once the verification results are available, the generated MCQs are refined and improved. This refinement process may involve rewording vague or awkwardly phrased stems, enhancing distractor quality, and discarding items that fail to meet clarity or validity standards [34], [131], [150], [164]. In most systems, this stage concludes with a human review by subject experts who assess the educational soundness and factual correctness of each question. After this review, questions are often arranged by difficulty level, formatted for delivery, and sometimes pilot-tested with a small group of learners to gather feedback before final deployment [2], [49]. Such a layered quality assurance process helps strike a balance between automation efficiency and the pedagogical rigor expected in real-world assessment settings.

VIII. EVALUATION METRICS

Evaluating automatically generated MCQs is not as simple as it may seem. It usually involves several complementary layers of review, some of which rely on algorithms and others that require human expertise. Researchers often focus on a few recurring factors: whether a question is relevant to the

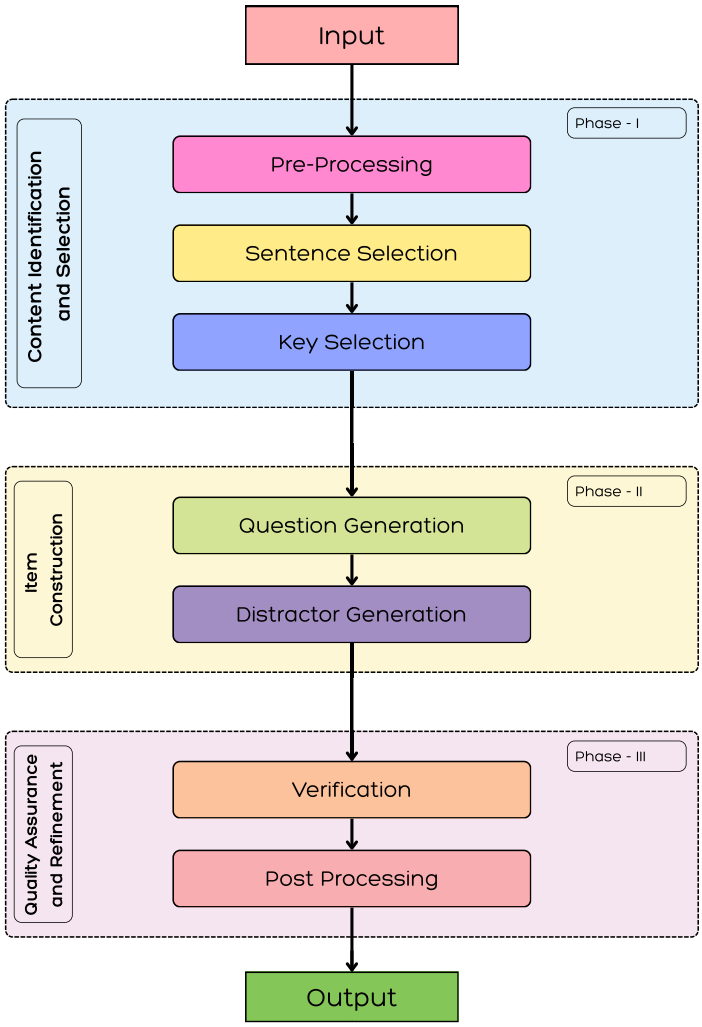


FIGURE 7. A generic architecture of MCQs generation process.

intended concept, how challenging it is for different learners, how convincing the distractors are, and whether the overall phrasing sounds natural and error-free. Table 22 provides a concise overview of the major approaches reported in the literature. Each method has its strengths but also certain drawbacks, depending on the learning context and the type of data used.

A. AUTOMATIC EVALUATION

Automatic evaluation focuses on measurable indicators that make large-scale assessment more practical. Instead of relying entirely on human raters, these methods compare generated questions or distractors with reference materials using similarity metrics. For instance, measures such as ROUGE-L or cosine similarity can be applied to gauge how closely the generated question aligns with the source text [96]. Precision-based measures like P@k [35] are also used to assess distractor quality, while some studies adopt information retrieval frameworks such as Retrieval-based Accuracy Differential (RAD) to evaluate the discriminative power of the generated items.

However, purely automated metrics often miss deeper educational value. A question may score high on textual similarity yet fail pedagogically by being too simple, ambiguous, or disconnected from the intended learning outcome [10], [18].

B. HUMAN EVALUATION

Human evaluation remains a cornerstone for assessing qualities that automated systems cannot capture well. Subject matter experts typically use rubrics, Likert scales, or qualitative commentary to assess fluency, instructional relevance, and distractor plausibility [106]. When paired with student response data, expert feedback can also help estimate item difficulty and discrimination. Yet, this process is labor-intensive, time-consuming, and occasionally

TABLE 21. Techniques for distractor generation.

Technique	How It Works	Example
Parts-of-Speech	Distractors must be the same grammatical type (e.g., all **nouns**).	"Respiration", "transpiration" (all nouns).
Frequency Matching	Uses distractors with a similar usage **frequency** to the correct answer.	"Meiosis", "mitosis" (commonly used terms).
WordNet/ Semantic	Generates **synonyms** or words with related meaning.	"Chemosynthesis", "autotrophy" (related biological processes).
Domain Ontology	Uses a **domain knowledge base** for highly specific, related terms.	"Chlorophyll" or a specific type of "cellular respiration".
Distributional Hypothesis	Identifies words that appear in the **same context**.	"Cellular respiration" or "glucose production" (contextually related).
Semantic Analysis	Creates related but **contextually inappropriate** options (often a common misconception).	"Respiration" (a distinct process).

inconsistent, as inter-rater agreement can vary despite clear guidelines [100].

C. PSYCHOMETRIC ANALYSIS

Psychometric analysis offers a quantitative perspective on question quality, particularly when learner response data is available. Using frameworks such as Classical Test Theory (CTT) or IRT, researchers can estimate item difficulty and discrimination indices and analyze distractor performance [104], [139]. These insights help determine whether a question behaves as expected in real testing scenarios. The downside is that psychometric validation requires responses from a sizable and representative learner population, which is rarely available during system design stages [132].

Across existing studies, there is still no universally accepted standard for evaluating MCQs generation. The selection of metrics often depends on the task, subject area, and language. A technique that works well for factual English questions may not suit reasoning-oriented or multilingual datasets [75], [96]. Consequently, many researchers recommend hybrid evaluation strategies that combine the scalability of automated methods with the contextual awareness of human reviewers and the empirical rigor of psychometric analysis. Such an integrative approach provides a more balanced and credible assessment of automatically generated questions.

IX. APPLICATIONS AND CASE STUDIES

In real-world applications where assessments play a key role in evaluations, MCQs have been used as a primary source due to their objective nature and ease of grading. Educational domains involve a lot to do with evaluations, and they are not the only domain that has adopted this method, but many industries are also using MCQs for skill assessments, hiring evaluations, and feedback gathering. We will discuss some of the most prominent applications in this section in order to find out how these systems are providing assistance in strengthening assessment and learning.

A. EDUCATION

One of the widely used fields where MCQs are the primary source for evaluation is education. In higher education, automated systems are used to generate questions for entrance assessment tests and even for classroom quizzes. Additionally, online learning platforms such as MOOCs have also integrated tools that are able to conduct short quizzes for each lecture. The primary benefit of these systems is the ability to provide feedback in a short period of time and help instructors supervise student progress at a massive scale, which would not be possible with manual work [6], [17], [42], [135], [174]. A recent example can be seen in some institutes where LLMs like GPT-4 have been used to create personalized questions in computer science courses that are adaptive to different learning styles [13], [47]. This allows evaluators to assess students' skills and learning abilities in a multi-dimensional way, providing a more in-depth analysis. Besides this, multi-stage prompting techniques have also been used to produce language-agnostic MCQs that broaden the reach of these tools in international education [95].

In the same way, MCQs are extensively employed in K-12 classrooms in order to measure the performance of students in a number of subjects such as mathematics, science, language, and arts [49], [53]. Many institutes are immediately adopting the tools that can automate the generation process in view of the fact that they hold the strength to enable the creation of personalized questions that directly aligned with curriculum standards [47], [88]. For example, some studies have discussed the implementation of certain tools like RevUP to evaluate the potential of automated MCQs generation systems, which have the ability to generate self-assessment questions in biology to reinforce student learning [100], [162]. Some case studies have reported the use of certain systems that produce MCQs directly from middle school textbooks to minimize the creation time for quizzes [44].

The domain of medical and healthcare depends greatly on MCQs to evaluate for high-stakes exams such as the United States Medical Licensing Examination (USMLE) [2], [10], [101], [181]. Clinical knowledge, diagnostic reasoning, and preparedness are some major areas where evaluations are required to get insights regarding students. These steps are required to help in making decisions with respect to curriculum design and difficulty adjustment for students. Recent efforts include using ChatGPT to generate pharmacotherapy questions with high-quality distractors [139] and employing AI to improve question quality for nursing education, enhancing the educational value of assessments [99].

B. PROFESSIONAL TRAINING AND LANGUAGE LEARNING

Over the last two decades, there has been a strong trend toward certifications in professional and corporate training programs. It can be observed in various fields comprising

TABLE 22. Summary of MCQs evaluation methods.

Evaluation Method	Type	Focus	Metrics	Strength	Limitations
Human/Expert Review [9], [36]–[38], [148]	Human	Overall quality, flawlessness, correctness	Rubrics, Likert scales, Binary (Accept/Reject)	Gold standard; Captures nuance.	Subjective, slow, expensive, unscalable.
Linguistic Review [1], [174]	Human	Grammatical correctness, fluency	Revision scales, Binary ratings	Validates text quality; Removes grammatical clues.	Highly dependent on rater expertise.
Semantic Review [57], [185]	Human	Clarity, coherence, contextual relevance	Likert scales, Binary relevance	Ensures grounding in source; Aligns with LOs.	Subjective; Can reflect underlying KB errors.
Inter-rater Agreement [92], [149], [165]	Human	Consistency of human ratings	Gwet's AC1, Fleiss' K, Cohen's Kappa	Quantifies reliability of human judgment.	Some metrics (Fleiss' K) weak with dominant labels.
Psychometric Analysis [75], [77], [150]	Hybrid	Item difficulty (ID), discriminating power (DP)	ID (Rasch), DP (Point-biserial), Distractor stats	Empirical proof of assessment value from real students.	Requires a student cohort; Not for real-time filtering.
LLMs Error Analysis [2], [10], [185]	Hybrid	LLMs failure modes (repetition, hallucination)	Exact match, Frequency counts, Beta regression	Pinpoints model weaknesses for targeted refinement.	Requires extensive calibration and complex stats.
N-gram Overlap [4], [82]	Automatic	Lexical/Syntactic similarity to references	BLEU, ROUGE-N, ROUGE-QAG	Automated, cheap, scalable, standard benchmark.	Poor correlation with semantic correctness.
Semantic Similarity [93], [105]	Automatic	Conceptual closeness, distractor plausibility	Cosine Sim (Embeddings), BERTScore, WMD	Measures meaning beyond word overlap.	Dependent on underlying embedding model quality.
Extractive Span Eval [152]	Automatic	Accuracy of extracted answer spans	Exact Match (EM), F1, Partial Match (PM)	Objective correctness measure for QA components.	EM is overly strict; penalizes minor phrasing variants.
Paraphrasing Metrics [10], [11], [185]	Automatic	Syntactic divergence from source text	METEOR, ROUGE-L, CIDER	Assesses novelty beyond exact word match.	Inverse interpretation needed for divergence tasks.
Complexity Classification [1], [18]	Automatic	Cognitive difficulty level (easy, medium, hard)	Probability distribution over difficulty classes	Automated difficulty scoring for adaptive testing.	Requires human-annotated training data.
Diversity Metrics	Automatic	Variety and non-redundancy of questions	Entropy (H(Q)), Self-BLEU	Quantifies the breadth of generated questions.	Relies on predefined question type categories.
Auxiliary Task Eval [18]	Hybrid	Answerability, groundedness, reasoning	Joint-EM, Joint-F1, Answerability uncertainty score	Validates question is answerable from context.	Conflates QG failure with comprehension failure.
Retrieval Ranking [59], [165]	Automatic	Relevance of retrieved context/passages	AUC-ROC, NDCG, MRR, P@k	Measures effectiveness of the underlying IR system.	Evaluates retrieval, not final MCQs quality.
Indistinguishability Test [2]	Human	Perceived human-likeness of questions	Binary (Human/Machine), Likert scale	Assesses naturalness and fluency of output.	Highly subjective and perception-based.
Internal Classifier Acc. [3], [8]	Automatic	Efficacy of pipeline components (e.g., filters)	Precision, Recall, Accuracy	Validates specific stages in the QG pipeline.	Dependent on the classifier's training data.
Affective State Assess. [9], [10]	Hybrid	Learner confusion/confidence via behavior	Correctness, time-per-Q, option changes	Captures behavioral signals beyond accuracy.	Requires specialized e-learning data collection.

of IT, finance, and healthcare, where MCQs are a standard assessment tool [83]. Automated systems can streamline the creation of relevant and challenging items. For instance, the AGenT Zero system uses zero-shot learning to generate skill assessment questions with minimal domain-specific input. This allow the system to support the integrity of certification processes [93].

Similarly, in language learning domain, it is a common practice to evaluate students in multiple aspects, such as grammar, vocabulary, and reading comprehension [72], [143], [145], [159], [174]. MCQs are frequently used as a primary question format which supports the idea of language learning using assessments. Automated MCQs generation systems can immediately provide targeted exercises to meet specific learning goals. Examples include a system that leverages GPT-3.5 Turbo to generate vocabulary-focused questions [153], and another that creates coherent quizzes from Turkish educational texts, demonstrating adaptability across languages [96].

C. ADAPTIVE LEARNING SYSTEMS

At the beginning, learning platforms were static in nature, which led toward repetition of questions making it easy for a student to pass the exam in multiple attempts. Adaptive learning platforms emerged as a groundbreaking solution to create dynamic and adaptive questions based on students' past performance [48], [88], [174]. These systems make use of the capabilities of automated MCQs generation systems to alter the parameters of assessments as a means to add

variations. Recently, researchers put forward a streamlined cloze test framework that dynamically adjusts the difficulty of each question and generate questions according to learner proficiency [88], [174]. By continuously monitoring performance, these systems adapt in real time to ensure that learners remain appropriately challenged, promoting individualized pacing and more effective learning.

X. DISCUSSION AND FUTURE DIRECTIONS

Over time, automated MCQs generation has developed progressively by utilizing various techniques introduced by researchers. Tasks that previously required extensive manual work can now be accomplished with improved efficiency and on a much larger scale. Recent advancement in generative AI has boosted the development of these systems by taking advantage of the competencies of state-of-the-art LLMs. Modern LLMs-based systems are capable of producing grammatically correct and contextually relevant questions [13], [77], [149], [181]. Despite the progress achieved so far, there is still room for improvement in areas that call for closer examination, particularly when it comes to maintaining educational depth, ensuring consistency, and addressing ethical concerns. The following discussion considers these issues and explores directions for future improvement.

A. PEDAGOGICAL QUALITY AND COGNITIVE DEPTH

While LLMs have greatly improved the fluency and readability of generated question stems [18], [149], they still struggle to test genuine understanding. The generated questions largely focus on recall and do not capture the technical aspects of learning that support reasoning and problem-solving competencies [186]. Besides, factual inaccuracies remain a common challenge for researchers while generating questions for technical subjects where precision is crucial [2], [10], [18], [49], [141], [181].

Improving factual reliability and ensuring that questions align with clear learning objectives should therefore be a central goal. One promising direction is to combine LLMs with external knowledge bases or lightweight fact-checking modules to minimize such errors [13], [26]. At the same time, enhancing the ability of NLP models to handle complex ideas and multi-step reasoning is equally important. Doing so would help generate questions that reflect a deeper, more conceptual level of understanding [4], [6].

B. CHALLENGE OF DISTRACTOR GENERATION

Designing meaningful distractors remains a difficult task. Although LLMs can generate grammatically sound options, they often miss the subtle misconceptions that learners commonly have [16], [18], [19]. One possible way forward is to involve teachers directly in the process, allowing models to learn from examples of real student mistakes [1], [16]. Future research might also focus on improved ranking or selection mechanisms that identify distractors that are not only plausible but also pedagogically useful [6].

C. STANDARDIZED EVALUATION AND BENCHMARKING

A consistent evaluation framework for automatically generated MCQs is still missing [90], [148]. Current evaluation relies either on automatic text-based metrics like ROUGE, which do not reflect learning value, or on human judgments, which are subjective and time-consuming. There is a clear need for shared benchmark datasets and transparent evaluation protocols [1]. Such frameworks should assess not only textual similarity but also the question's difficulty, cognitive level, and educational relevance [2], [4], [13], [46], [91]. Establishing reporting standards would also make it easier to compare findings and track real progress across studies.

D. ETHICAL CONSIDERATIONS AND MITIGATING BIAS

Given the fact that large language models (LLMs) are trained on large amounts of web-based text data, they are sometimes prone to reproduce social or cultural biases [186]. Such biases may lead to content that is misleading or inequitable, which is a particular problem in the educational domain. Inadequately constructed distractors can also confuse learners or distort the results of the assessment. As such, to address these concerns, it is necessary to have transparent auditing procedures and strong strategies to mitigate bias [10], [176]. Establishing clear ethical principles for deploying AI in educational settings will help ensure fairness, accountability, and trust.

In order to illustrate how these ethical issues play out in practice, one might look at a few cases. In standardized medical licensing examinations, case vignettes developed using LLMs have been noted to have over-representation of certain demographic characteristics and under-representation of others, which may be prejudicial towards certain groups of students during assessment [10], [176]. On another front, educational platforms that accumulate student response data to increase the efficiency of question generators must work through data protection laws including the Family Educational Rights and Privacy Act (FERPA) and General Data Protection Regulation (GDPR) and raise important questions about consent, data retention, and limitation of use.

Furthermore, there is the problem of academic integrity: the availability of practice questions generated by LLM-based generators will become so widespread that students can create similar questions to the real ones in the examination, undermining the validity and security of high-stakes examinations. Empirical evidence further suggests that even though a large proportion of LLM-generated multiple-choice questions are explicitly prompted for higher-order thinking questions consistent with Bloom's Taxonomy, a large proportion of LLM-generated questions are for lower levels of cognition (knowledge and comprehension) rather than analysis or evaluation [175], [186]. These observations serve to highlight the need for rigorous systems for reviewing the use of such tools before deploying them into actual educational settings. These examples highlight the need for stringent validation procedures and human oversight before

LLM-generated assessments can be widely implemented in real educational contexts.

E. HUMAN–AI COLLABORATION

Even with the growing level of automation, the knowledge of educators still plays an indispensable role [46], [91], [181]. AI tools should be conceived as assistance systems instead of replacement systems. Human supervision remains crucial for ensuring accuracy, clarity, and pedagogical validity [49]. Future research could thus concentrate on creating platforms with interactive features in supporting teachers to refine and validate AI-generated content [1] in order to combine human insight and computational efficiency.

F. CRITICAL GAPS IN LLM-BASED METHODS

While the preceding sections have characterized general challenges, it is necessary to consider critically the limitations of specific approaches based on LLMs that make the present ones different from the previous methods. First, the topic of factual reliability stands out because in opposition to rule-based systems which manage to generate content only from verified resources, LLMs are usually inclined to present plausible statements but erroneous factually, which is especially harmful in areas such as medicine, law, and engineering, where accuracy is non-negotiable [10], [49], [141]. Second, in terms of creating distractors, LLMs will tend to choose options based on surface-level word associations and not based on genuine learner misconceptions, whereas the classical approaches, which use ontologies or learner error databases, can provide more pedagogically meaningful alternatives [16], [19]. Third, difficulty control is unreliable even with explicit prompting there is unpredictable variance in the cognitive level of the items generated, which makes it difficult to create balanced assessments [88], [175]. Fourth, while rule-based and ontology-based systems have transparency in question generation, LLMs are opaque and it is hard for educators to understand the reasoning behind an individual question or systematically improve the output.

Finally, there is a disconnect between current evaluation metrics and pedagogical goals. Automatic evaluation metrics like BLEU and ROUGE evaluate textual similarity but not the effectiveness of instruction, meaning it is possible for a system to achieve a high score on evaluation benchmarks while scoring low on teaching effectiveness [90], [148]. Acknowledging these gaps is essential in helping to guide future research efforts towards creating solutions that strike a balance between the generative abilities of LLMs and the reliability and precision of previous methodologies. It is important to recognize these gaps to inform future research and development, aiming to combine the generative capabilities of LLMs with the reliability and precision of past methodologies.

G. EMERGING RESEARCH FRONTIERS

Looking ahead, there are several directions worth exploring. One is to design systems that perform well across multiple subjects and educational levels without needing heavy domain-specific data [14], [23], [96], [102]. Another is adaptive learning, where the system adjusts question difficulty to match a learner's progress [5], [6], [8]. Work on prompt design for complex STEM disciplines is also essential [17], [18]. Finally, instead of generating isolated questions, researchers should aim to create cohesive assessments or multi-step problems that better reflect how real understanding develops [13].

XI. CONCLUSION

The field of automated MCQs generation has evolved substantially over the years by taking advantage of state-of-the-art research technologies. In the past, these systems relied on fixed, rule-based templates, but recent advancement in AI and the availability of strong LLMs have totally changed the landscape substantially. They are capable of generating high quality text from simple prompting techniques and are able to produce questions that sound more natural looking and reflect context with improved accuracy. In this survey, we explored how the field has evolved from basic template-matching techniques, outlined a general framework, and discussed the main datasets and evaluation practices that have guided current research.

Despite these improvement particularly in question generation, there are a number of key challenges that still remain in this domain. While LLMs can create fluent and contextually relevant questions, they frequently fall short in capturing true educational depth or aligning closely with specific learning goals. Some recent studies have shown that many of the questions generated by these systems have mainly focused on simple recall rather than deeper reasoning or critical thinking. Another persistent issue is generating multiple options designed to distract test takers so that these options feel convincing and reflect genuine student misconceptions.

In order to improve this field, it is necessary to address these existing challenges. Researchers should focus their future work on improving domain-specific adaptability and developing more effective techniques for distractor generation. Similarly, consistent and reliable evaluation standards are equally important for assessing not only language quality but also educational value. Overall, the goal should be to move beyond simple automation and towards systems that genuinely assist teachers and learners. By combining the strengths of AI with educational insight, we can work towards assessments that are not only efficient to create but also meaningful and sound in how they measure learning.

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grammar throughout the manuscript. All intellectual contributions, analysis, and conclusions remain solely those of the authors.

LIST OF ABBREVIATIONS

AI	Artificial Intelligence.
AQG	Automatic question generation.
BART	Bidirectional and Auto-Regressive Transformers.
BERT	Bidirectional Encoder Representations from Transformers.
BLEU	Bilingual Evaluation Understudy.
CAT	Computerized Adaptive Test.
CoT	Chain-of-Thought.
CTT	Classical Test Theory.
DG	Distractor Generation.
DL	Deep Learning.
GPT	Generative Pre-trained Transformer.
GRE	Graduate Record Examination.
ID	Item Discrimination.
IDF	Inverse Document Frequency.
IRT	Item Response Theory.
LLMs	Large Language Models.
LSTM	Long Short Term Memory.
MCQs	Multiple Choice Questions.
ML	Machine Learning.
MOOCs	Massive Open Online Courses.
NBME	National Board of Medical Examiners.
NER	Named Entity Recognition.
NLP	Natural Language Processing.
OWL	Web Ontology Language.
PaLM	Pathways Language Model.
POS	Part-of-speech.
QG	Question Generation.
RAD	Retrieval-based Accuracy Differential.
RAG	Retrieval-Augmented Generation.
RNNs	Recurrent Neural Networks.
ROUGE	Recall-Oriented Understudy for Gisting Evaluation.
ROUGE-L	Recall-Oriented Understudy for Gisting Evaluation-Longest Common Subsequence.
SAQs	Short-Answer Questions.
SAT	Scholastic Assessment Test.
SRL	Semantic Role Labeling.
SVM	Support Vector Machine.
SVO	Subject-Verb-Object.
T5	Text-to-Text Transfer Transformer.
TF-IDF	Term Frequency-Inverse Document Frequency.
TOT	Tree-of-Thought.
USMLE	United States Medical Licensing Examination.
VSAQs	Very Short-Answer Questions.
WSD	Word Sense Disambiguation.

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